



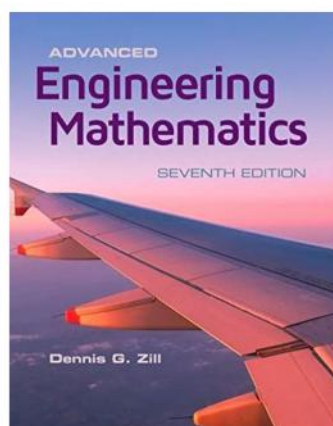
Get sticker for
calculator!!!!!!

ENGR 213-Ordinary Differential Equations

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Week one Lectures:

- Introduction to differential equations
- Section 1.1 and 1.2 (1.3) of textbook
- Section 2.1



Introduction

- A differential equation is an equation in which one or more dependent variable (or function), and one or more derivative of that variable (or function) over an independent variable, occurs.

- ODE vs PDE

$$\frac{dy}{dx} + y = 0$$

$$\left(\frac{dy}{dx}\right)^2 + 2xy = \cos(x) \quad \text{Ordinary differential equation}$$

$$x(x,t) \quad \frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 \theta}{\partial t^2} = 0$$

- Notation: We will use different notation to represent a derivative

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, y', y''$$

$$\frac{dy}{dt} = \dot{y} \quad \text{dot notation for time}$$

$$\frac{d^2y}{dt^2} = \ddot{y}$$

- Order:

The order of an ODE is the order of the highest derivative in the equation.

$$\frac{dy}{dx} - y^4 = 0 \rightarrow 1^{\text{st}} \text{ order}$$

$$\left(\frac{dy}{dx}\right)^3 + y = 0 \rightarrow 1^{\text{st}} \text{ order}$$

$$\frac{d^4y}{dx^4} + \frac{d^2y}{dx^2} + 1 = 0 \Rightarrow 4 \text{ order}$$

$$\uparrow \quad \uparrow$$

4 order > 2 order

- Linearity: an ODE is linear if
 - all dependent y , and derivative, are of power 1
 - all coefficient multiplying those y are, at most, function of independent x

$$a_n(x) \frac{d^{(n)}}{dx^n} y(x) + a_{n-1}(x) \frac{d^{(n-1)}}{dx^{n-1}} y(x) + \dots + a_1(x) \frac{dy}{dx} + a_0(x) y = g(x)$$

↑

- If not linear, non-linear...

- Homogeneous:

An ODE is said homogeneous if the value of $g(x)$ is 0. If not, the ODE is said non-homogenous

- A solution of a given differential equation of order n on some open interval I is a function ϕ that has at least n derivative (continuous) on the interval I .

- Explicit: $y = \phi(x)$

- Implicit: $G(y,x) = 0$

Have to write or lose points in exam
 $x^2 + y^2 = 4$

- Different types of solution:

general, particular, singular, trivial

Domain $\rightarrow$ "Where" the solution is valid
 $\rightarrow$ limitation for equation
 $\frac{dy}{dx} + \frac{1}{x} y = 0$
 $\rightarrow$ Solution continuous

All possible solⁿ

$y' = \cos(x)$
 $\rightarrow y = \sin(x) + C$

$C=3$
 $y = \sin(x) + 3$
 $\rightarrow$ no unknown left

$\rightarrow$ special solution

$\rightarrow 0$

- Initial-Value Problems (IVP)

main topic
this semester

If n values for $y(x)$ and its $n-1$ derivative are given at one point (x_0) for the solution $y(x)$ of an n th order ODE, this ODE is an Initial Value Problem (IVP) and $y(x_0), y'(x_0), \dots$ are the initial conditions.

- Boundary Value Problem (BVP)

If the solutions of an ODE are given at several points $(x_0, x_1, x_2, \dots)$, then the problem is called a Boundary Value Problem (BVP)

Ex: If $x(t) = c_1 \cos(4t) + c_2 \sin(4t)$ is a solution of $x'' + 16x = 0$, find the solution of the IVP with $x(\frac{\pi}{2}) = -2$ and $x'(\frac{\pi}{2}) = 1$

$$x(\frac{\pi}{2}) = c_1 \cos(\underbrace{4 \cdot \frac{\pi}{2}}_{2\pi}) + c_2 \sin(\underbrace{4 \cdot \frac{\pi}{2}}_{2\pi}) = c_1 + c_2 \cdot 0 = c_1 = -2$$

$$x' = -2 \cdot (-4 \sin(4t)) + c_2 \cdot 4 \cos(4t)$$

$$x'(t) = 8 \sin(4t) + 4 c_2 \cos(4t)$$

$$x'(\frac{\pi}{2}) = 4 c_2 = 1 \rightsquigarrow c_2 = \frac{1}{4}$$

sol: $x(t) = -2 \cos(4t) + \frac{1}{4} \sin(4t)$

Existence and Uniqueness theorem

$$\frac{dy}{dx} = f(x, y) \quad , \quad y(x_0) = y_0$$

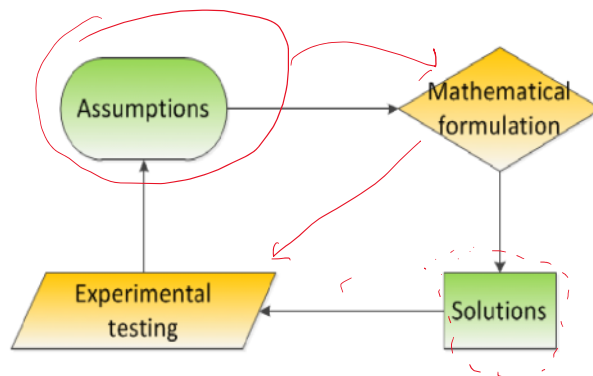
THEOREM 1.2.1 Existence of a Unique Solution

Let R be a rectangular region in the xy -plane defined by $a \leq x \leq b$, $c \leq y \leq d$, that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f / \partial y$ are continuous on R , then there exists some interval $I_0: (x_0 - h, x_0 + h)$, $h > 0$, contained in $[a, b]$, and a unique function $y(x)$ defined on I_0 that is a solution of the initial-value problem (2).

Why ODE?

- Mathematical modeling of the world!

$$\frac{dT}{dt} \propto \Delta T$$



Chapter 2: First order ODE

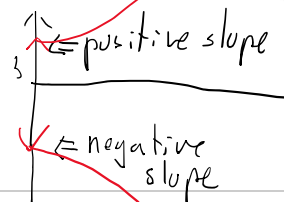
- This chapter will focus on the different technique to solve different kind of first order ODE.
- A lot of different techniques, so don't get behind!

2.1: Geometric introduction

Napkin solution

- Understanding the geometry of an equation will help in some case to have an idea of the solution
- Definition:
 - An autonomous equation is an equation of the form $\frac{dy}{dx} = f(y)$
 - A real number c is called a critical point of an autonomous equation if it is a zero of $f(y)$ ($f(c)=0$): it is an equilibrium point!
 - C is a constant solution of the equation!
- As $\frac{dy}{dx}$ change sign at c , we can estimated the change in the solution

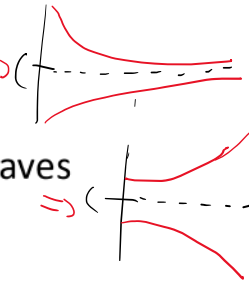
ex: $f(y) = (y-3) \Rightarrow f(3) = 0$
 $f(2) = -1$ $3 \Rightarrow$ critical point



$C =$ critical point

- Attractor: a point where the slope lines goes $\Rightarrow$

- Attractor: a point where the slope lines goes $\Rightarrow$
 - stable
- Repeller: a point from where the slope lines leaves $\Rightarrow$
 - Unstable
- If both: a saddle point or a semi-stable point



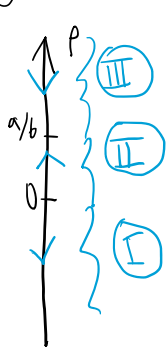
- Translation: If $y(x)$ is a solution of an autonomous ODE, then $y(x) = y(x-k)$ is also a solution, provided k is a constant

example: Study the behaviour of $\frac{dP}{dt} = P(a-bP)$, $a > 0$, $b \geq 0$
Fill the phase portrait

① $\frac{dP}{dt} = P(a-bP) \rightarrow$ autonomous $\checkmark$

$$\begin{aligned} a - Pb &= 0 \\ a &= bP \\ P &= \frac{a}{b} \end{aligned}$$

② Find critical point: $P(a-bP) = 0$ $\boxed{P=0}$



③ region I $-\infty < P < 0$ $P \ominus$

$$P(a-bP) < 0$$

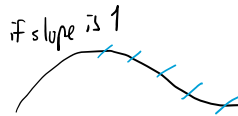
II $0 < P < \frac{a}{b}$ $P \oplus$ $P < \frac{a}{b} \Rightarrow bP < a$
 $(a-bP) > 0$

III $\frac{a}{b} < P < \infty$ $\frac{a}{b} < P \Rightarrow a < Pb$
 $(a-bP) < 0$

Direction field (slope field)

- A method to visualize the solution:

- ODE in the explicit form: $\frac{dy}{dx} = f(x,y)$
- For this procedure, it is not important if it is autonomous or not (see phase portrait in your book for a different method)
- Assume $f(x,y)$ is constant
- Draw isoclines
- Draw slope line along the isocline
- Approximate solution
- See example..



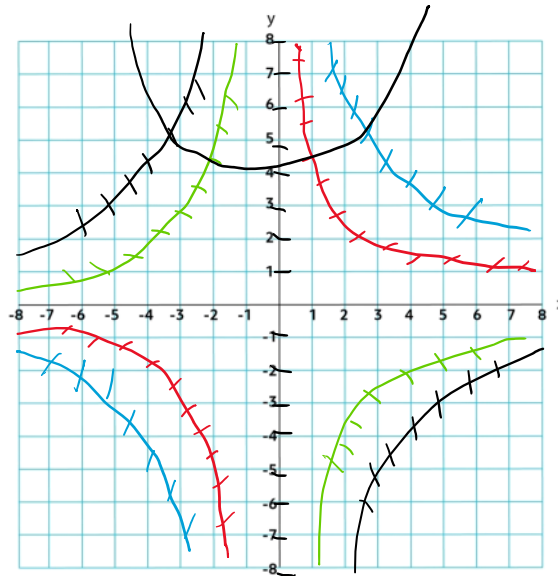
Ex: $y'(x) = xy$

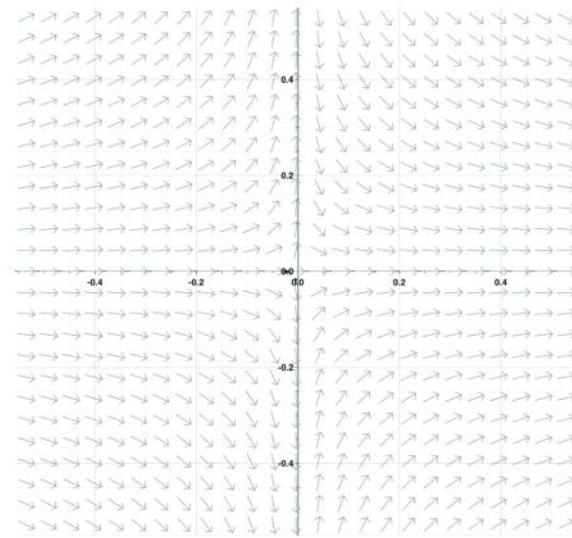
① isocline

→ $xy = \text{cst}$

$xy = 0$
 $xy = 1$ 45°
 $xy = -1$ 135°
 $xy = 2$
 $xy = -2$

② slope element





1.1 Q13 $\underbrace{2y' + y = 0}_{\text{ODE}}, \underbrace{y = e^{-x/2}}_{\text{eq}}, y' = -\frac{1}{2}e^{-x/2}$

$$\rightarrow 2\left(-\frac{1}{2}e^{-x/2}\right) + e^{-x/2} = -e^{-x/2} + e^{-x/2} = 0$$

Therefore, $y = -\frac{1}{2}e^{-x/2}$ satisfies the differential equ.

$\rightarrow$ suitable interval $\rightarrow I = (-\infty, +\infty)$

Q17 $\underbrace{(y-x)y'}_{\text{I}} = \underbrace{y-x+8}_{\text{II}}, y = x+4\sqrt{x+2}, y' = 1+4\left(\frac{1}{2\sqrt{x+2}}\right) = 1 + \frac{2}{\sqrt{x+2}}$

$$(y-x)y' = (x+4\sqrt{x+2}-x)\left(1+\frac{2}{\sqrt{x+2}}\right) = 4\sqrt{x+2}+8 \quad (\text{I})$$

$$y-x+8 = x+4\sqrt{x+2}-x+8 = 4\sqrt{x+2}+8 \quad (\text{II})$$

$$(\text{I}) = (\text{II}) \quad \phi = x+4\sqrt{x+2}, x \geq -2 \rightarrow I = [-2, +\infty)$$

$$(x+2)^{\frac{1}{2}} \rightarrow \frac{1}{2}(x+2)^{-\frac{1}{2}} \rightarrow \frac{1}{2(x+2)^{\frac{1}{2}}}$$

Q25 $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0, y = C_1 e^{2x} + C_2 x e^{2x}$
 $\hookrightarrow y = e^{2x}(C_1 + C_2 x)$

$$y' = 2e^{2x}(C_1 + C_2 x) + e^{2x}(C_2)$$

$$y' = e^{2x}(2C_1 + C_2 + 2C_2 x)$$

$$y'' = 2e^{2x}(2C_1 + C_2 + 2C_2 x) + e^{2x}(2C_2)$$

$$y'' = e^{2x}(4C_1 + 4C_2 + 4C_2 x)$$

$$\rightarrow y'' - 4y' + 4y = 0$$

$$\rightarrow e^{2x}(4C_1 + 4C_2 + 4C_2 x) - 4e^{2x}(2C_1 + C_2 + 2C_2 x)$$

$$\rightarrow e^{2x}[(4C_1 + 4C_2 x) - (8C_1 + 4C_2 + 8C_2 x) + (4C_1 + 4C_2 x)] = 0 \rightarrow e^{2x}[0] = 0$$

$$I = (-\infty, +\infty)$$

2.1

In Problems 21–28, find the critical points and phase portrait of the given autonomous first-order differential equation. Classify each critical point as asymptotically stable, unstable, or semi-stable. By hand, sketch typical solution curves in the regions in the xy -plane determined by the graphs of the equilibrium solutions.

21. $\frac{dy}{dx} = y^2 - 3y$

22. $\frac{dy}{dx} = y^2 - y^3$

Q21 $\frac{dy}{dx} = y^2 - 3y$

$$y^2 - 3y = 0 \rightarrow y(y-3) = 0 \rightarrow y = 0, 3$$

Interval	$f(y) = y(y-3)$	Direction
$y < 0$	+	$y \nearrow$
$0 < y < 3$	-	$y \searrow$
$y > 3$	+	$y \nearrow$



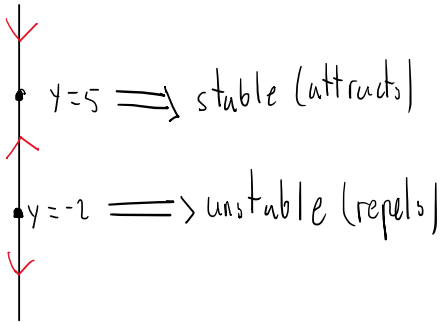
$y=0 \rightarrow$ it is asymptotically stable
 $y=3 \rightarrow$ it is unstable

Q 24. $\frac{dy}{dx} = 10 + 3y - y^2$

$$\frac{dy}{dx} = 10 + 3y - y^2 \quad 10 + 3y - y^2 = 0 \rightarrow -(y - 3y - 10) = 0$$

$$\rightarrow -(y - 5)(y + 2) = 0 \rightarrow (5 - y)(y + 2) = 0$$

$$y = -2, 5$$



Interval	$f(y) = (5-y)(y+2)$	Direction
$y < -2$	-	$y \searrow$
$-2 < y < 5$	+	$y \nearrow$
$y > 5$	-	$y \searrow$

2.2

In Problems 1-22, solve the given differential equation by separation of variables.

$$\frac{dy}{dx} = g(x)h(y) \rightarrow \int \frac{dy}{h(y)} = \int g(x) dx$$

1. $\frac{dy}{dx} = \sin 5x$

2. $\frac{dy}{dx} = (x+1)^2$

3. $dx + e^{3x} dy = 0$

Q3 $dx + e^{3x} dy = 0 \rightarrow \int dy = \int -e^{-3x} dx$

$$\rightarrow y = \frac{1}{3} e^{-3x} + C$$

6. $\frac{dy}{dx} + 2xy^2 = 0$

Q6 $\frac{dy}{dx} + 2xy^2 = 0 \rightarrow \frac{dy}{dx} = -2xy^2 \rightarrow \int \frac{dy}{y^2} = \int -2x dx$

$$\rightarrow -\frac{1}{y} = -x^2 + C \rightarrow \frac{1}{y} = x^2 + C$$

$$\rightarrow y = \frac{1}{x^2 + C}$$

20. $\frac{dy}{dx} = \frac{xy + 2y - x - 2}{xy - 3y + x - 3}$

Q20 $\frac{dy}{dx} = \frac{(xy + 2y - x - 2)^*}{(xy - 3y + x - 3)^{**}}$

$$* (xy + 2y) + (-x - 2) = y(x+2) - (x+2)$$

$$= (x+2)(y-1)$$

$$** (xy - 3y) + (x - 3) = y(x-3) = y(x-3) + (x-3)$$

$$= (x-3)(y+1)$$

$$\frac{dy}{dx} = \frac{(x+2)(y-1)}{(x-3)(y+1)} \rightarrow \int \left(\frac{y+1}{y-1} \right) dy = \int \left(\frac{x+2}{x-3} \right) dx$$

$$\int \left(\frac{y-1+2}{y-1} \right) dy = \int \left(\frac{x-3+5}{x-3} \right) dx = \int \left(1 + \frac{2}{y-1} \right) dy = \int \left(1 + \frac{5}{x-3} \right) dx$$

$$(x+2)(y-1) = x+5 \quad (y-3)(y+1) = y-2$$

$$\int \frac{1}{|y-1|^2} dy = \int \frac{1}{(x-3)^2} dx = \int \left(1 + \frac{5}{y-1}\right) dy = \int \left(1 + \frac{5}{x-3}\right) dx$$

$$y + 2 \ln |y-1| = x + 5 \ln |x-3| + C$$

$$\rightarrow \frac{(y-1)^2}{(x-3)^2} = C_1 e^{x-y}$$

$$y - x - C = \ln |x-3|^5 - \ln |y-1|^2 = \ln \frac{|x-3|^5}{|y-1|^2} = y - x - C$$

$$\ln \frac{|y-1|^2}{|x-3|^5} = x - y + C$$

$$\rightarrow e^{x-y+C} = \frac{(y-1)^2}{(x-3)^5} \rightarrow C_1 e^{x-y} = \dots$$

- 1.2 In Problems 7–10, $x = c_1 \cos t + c_2 \sin t$ is a two-parameter family of solutions of the second-order DE $x'' + x = 0$. Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions.

7. $x(0) = -1, x'(0) = 8$

8. $x(\pi/2) = 0, x'(\pi/2) = 1$

9. $x(\pi/6) = \frac{1}{2}, x'(\pi/6) = 0$

10. $x(\pi/4) = \sqrt{2}, x'(\pi/4) = 2\sqrt{2}$

Q7 $x(0) = c_1 \cos(0) + c_2 \sin(0) = -1$

$c_1 \cdot 1 + c_2 \cdot 0 = -1$

$c_1 = -1$

$x'(0) = -c_1 \sin(0) + c_2 \cos(0) = 8$

$0 + c_2 \cdot 1 = 8$

$c_2 = 8$

Sol: $x = -\cos t + 8 \sin t$

Q9 $x(\frac{\pi}{6}) = c_1 \cos(\frac{\pi}{6}) + c_2 \sin(\frac{\pi}{6}) = \frac{1}{2}$

$c_1 \left(\frac{\sqrt{3}}{2}\right) + c_2 \left(\frac{1}{2}\right) = \frac{1}{2} \Rightarrow c_1 \sqrt{3} + c_2 = 1 \quad (1)$

$(2) \rightarrow (1)$

$x'(\frac{\pi}{6}) = -c_1 \sin(\frac{\pi}{6}) + c_2 \cos(\frac{\pi}{6}) = 0$

$\Rightarrow -c_1 \left(\frac{1}{2}\right) + c_2 \left(\frac{\sqrt{3}}{2}\right) = 0 \Rightarrow c_2 \left(\frac{\sqrt{3}}{2}\right) = c_1 \left(\frac{1}{2}\right) \Rightarrow c_1 = c_2 \sqrt{3} \quad (2)$

$(c_2 \sqrt{3})(\sqrt{3}) + c_2 = 1$

$c_2(3) + c_2 = 1$

$4c_2 = 1 \Rightarrow c_2 = \frac{1}{4}$

With $c_2 = \frac{1}{4} \rightarrow (2) \quad c_1 = \frac{1}{4} \sqrt{3} \Rightarrow c_1 = \frac{\sqrt{3}}{4}$

Sol: $\frac{\sqrt{3}}{4} \cos(t) + \frac{1}{4} \sin(t)$

In Problems 11–14, $y = c_1 e^x + c_2 e^{-x}$ is a two-parameter family of solutions of the second-order DE $y'' - y = 0$. Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions.

11. $y(0) = 1, y'(0) = 2$

12. $y(1) = 0, y'(1) = e$

13. $y(-1) = 5, y'(-1) = -5$

14. $y(0) = 0, y'(0) = 0$

$y' = c_1 e^x - c_2 e^{-x}$

Q11 $y(0) = c_1 e^0 + c_2 e^{-0} = 1$

$c_1 + c_2 = 1 \quad (1) \Rightarrow c_1 = 1 - c_2$

$y'(0) = c_1 e^0 - c_2 e^{-0} = 2$

$c_1 - c_2 = 2 \quad (2) \Rightarrow c_2 = -2 + c_1$

$c_1 = 1 - (-2 + c_1)$

$c_1 = 1 + 2 - c_1$

$2c_1 = 3 \Rightarrow c_1 = \frac{3}{2}$

With $c_1 = \frac{3}{2} \rightarrow (2) \quad c_2 = -2 + \frac{3}{2} = -\frac{1}{2}$

Sol: $y = \frac{3}{2} e^x - \frac{1}{2} e^{-x}$

Q12 $y(1) = c_1 e^1 + c_2 e^{-1} = 0$

$c_1 e + c_2 e^{-1} = 0 \Rightarrow (1) \quad (c_1 e + c_2 e^{-1}) + (c_1 e - c_2 e^{-1}) = 0 + e$

$y'(1) = c_1 e^1 - c_2 e^{-1} = e$

$c_1 e - c_2 e^{-1} = e \Rightarrow (2) \Rightarrow 2c_1 = 1$

$c_1 = \frac{1}{2}$

With $c_1 = \frac{1}{2} \rightarrow (2) \quad \frac{1}{2} e - c_2 e^{-1} = e \Rightarrow -c_2 e^{-1} = \frac{1}{2} e$

$\Rightarrow c_2 e^{-1} = -\frac{1}{2} e$ x both sides by e

$\Rightarrow c_2 = -\frac{1}{2} e^2$

Sol: $\frac{1}{2} c_1 e^x - \frac{1}{2} e^2 e^{-x}$

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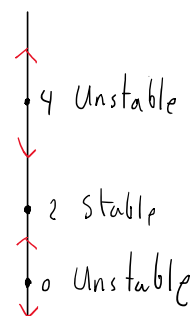
23. $\frac{dy}{dx} = (y - 2)^4$

24. $\frac{dy}{dx} = 10 + 3y - y^2$

25. $\frac{dy}{dx} = v^2(4 - v^2)$

Q26 Critical points are $y = 0, 2, 4$

interval	$y(2-y)(4-y)$	direction
$-\infty < y < 0$	$y = -1 \quad -$	$\searrow$
$0 < y < 2$	$y = 1 \quad +$	$\nearrow$
$2 < y < 4$	$y = 3 \quad -$	$\searrow$
$y > 4$	$y = 5 \quad +$	$\nearrow$



23. $\frac{dy}{dx} = (y-2)^4$

24. $\frac{dy}{dx} = 10 + 3y - y^2$

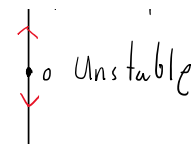
25. $\frac{dy}{dx} = y^2(4 - y^2)$

26. $\frac{dy}{dx} = y(2-y)(4-y)$

27. $\frac{dy}{dx} = y \ln(y+2)$

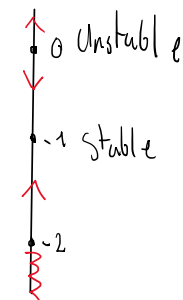
28. $\frac{dy}{dx} = \frac{ye^y - 9y}{e^y}$

$2 < y < 4$ $y=3$ $-$ $\searrow$
 $y > 4$ $y=6$ $+$ $\nearrow$



Q27 Critical points are $y=0, -1, -2 \rightarrow \ln(y+2)=0 \Rightarrow y+2=e^0 \Rightarrow y+2=1 \Rightarrow y=-1$
 (ln) domain $(-2, \infty)$

interval	Ex	sign $y \ln(y+2)$	direction
$-2 < y < -1$	-1.5	+	$\nearrow$
$-1 < y < 0$	-0.5	-	$\searrow$
$y > 0$	1	+	$\nearrow$





Sept 22 → quiz 1 ⇒ One question on 2.2 OR 2.3
Weekly Webworks are due on Sundays

Lecture 2

ENGR213

2.2 Separable Equations

~~$(x+y)^2$~~
 $\sin(x) \cos(x)$ ✓

- A differential equation in the form $\frac{dy}{dx} = g(x)f(y)$ is said separable

- Technique to solve:

① • Separate the variables: $\frac{dy}{f(y)} = g(x)dx$

- ② • Integrate:

$$\int \frac{dy}{f(y)} = \int g(x)dx + c$$

Solve $y' = ky$ if $k > 0$ and $y(0) = 4$

Examples:

① $-\frac{dy}{dx} = kx \Rightarrow$ separate $\boxed{\frac{dy}{y} = kdx}$

② $\int \frac{dy}{y} = \ln|y| \quad / \quad \int kdx = kx + C$

$\boxed{\ln|y| = kx + C} \Rightarrow$ explicit solution would be better

$\ln|y| = (kx + C) \Rightarrow y = e^{kx} \underbrace{e^C}_{C_1} \Rightarrow \boxed{y(x) = C_1 e^{kx}} \leftarrow \text{general}$

IVP $y(0) = 4 = C_1 e^0 \Rightarrow C_1 = 4 \Rightarrow \boxed{y(x) = 4e^{kx}} \leftarrow \text{particular}$

• Solve $9yy' + 4x = 0$ $\leftarrow$ implicit solution $\Rightarrow 9y \frac{dy}{dx} + 4x = 0$

$$9y \frac{dy}{dx} = -4x$$

$$\Rightarrow \boxed{9y dy = -4x dx}$$

② Integral $\Rightarrow \int 9y dy = - \int 4x dx$
 $\frac{9y^2}{2} = - \frac{4x^2}{2} + C$

$$\frac{9y^2 + 4x^2}{2} = C \rightsquigarrow \boxed{9y^2 + 4x^2 = 2C}$$

$$9y^2 + 4x^2 = C \times$$

$$9y^2 + 4x^2 = C_1 \checkmark$$

Solve $\cos(x) [e^{2y} - y] \frac{dy}{dx} = e^y \sin 2x$ for $y(0) = 0$.

① Separate: $\boxed{\left[\frac{e^{2y} - y}{e^y} \right] dy = \frac{\sin 2x}{\cos x} dx}$

* using $\sin 2x = 2 \sin x \cos x$, we have $\boxed{[e^y - y e^{-y}] dy = 2 \sin x dx}$

② Integrate: $\int e^y dy - \int y e^y dy = 2 \int \sin x dx + C$
 $\left[\begin{array}{l} u = y \quad \frac{du}{dy} = 1 \\ dv = e^y dy \quad v = e^y \end{array} \right]$
 $e^y - [y e^y - \int e^y dy] = 2[-\cos x] + C$
 $e^y + y e^{-y} + e^{-y} = -2 \cos x + C$

③ IVP: if $x=0, y=0 \Rightarrow 1 + 0 + 1 = -2 + C \therefore C = 4$

$\Rightarrow \boxed{e^y + e^{-y} [y + 1] = 4 - 2 \cos x}$

2.3 Linear Equations

- A first order differential equation is said to be linear if it can be written in the form:

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- The **standard form** is found by dividing this equation by $a_1(x)$, provided that it is not 0.

$\boxed{\frac{dy}{dx} + P(x)y = f(x)}$, provided $P(x) = \frac{a_0}{a_1}$ and $f(x) = \frac{g(x)}{a_1}$

- Two different technique whether the equation is **homogeneous** or not

$\frac{dy}{dx} + P(x)y = 0$ $\rightarrow$ Separable $\frac{dy}{dx} = -P(x)y$ $\frac{dy}{y} = -P(x) dx$ $y = C e^{-\int P(x) dx}$

- Homogeneous equation:

- Separation of variable!

$$y_c = C e^{-\int P(x) dx}$$

- Non-Homogeneous:

$$\frac{d}{dx} [y e^{\int P(x) dx}] = \frac{dy}{dx} \cdot e^{\int P(x) dx} + y \frac{d}{dx} [e^{\int P(x) dx}]$$

The solution is found from the sum of an homogeneous and an non-homogeneous solution, generally written as $y = y_c + y_p$

non homogeneous:

The solution is found from the sum of an homogeneous and an non-homogeneous solution, generally written as: $y = y_h + y_p$

Multiplying factor!

$$\Rightarrow e^{\int P(x) dx} \left[\frac{dy}{dx} + P(x) y \right] = e^{\int P(x) dx} f(x)$$

$$e^{\int P(x) dx} \left[\frac{dy}{dx} + y \left[P(x) e^{\int P(x) dx} \right] \right] = e^{\int P(x) dx} f(x)$$

$$\frac{d}{dx} \left[e^{\int P(x) dx} \right] = e^{\int P(x) dx} \cdot P(x)$$

$$\frac{d}{dx} [y e^{\int P(x) dx}] = e^{\int P(x) dx} f(x)$$

Separable

$$\frac{d[y e^{\int P(x) dx}]}{dw} = e^{\int P(x) dx} f(x) dx$$

$$w = y e^{\int P(x) dx}$$

Examples:

using linear equation tech.

• Solve $y' - 3y = 6$

$$\frac{dy}{dx} - 3y = 6$$

① Find $P(x) = -3$
 $f(x) = 6$

② $e^{\int P(x) dx} = e^{\int (-3) dx} = e^{-3x}$

③ $\frac{d}{dx} [y e^{\int P(x) dx}] = e^{\int P(x) dx} f(x)$

$$\frac{d}{dx} [y e^{-3x}] = e^{-3x} 6$$

④ Separate $\Rightarrow d[y e^{-3x}] = 6 e^{-3x} dx$

$$y e^{-3x} = 6 \cdot \left(-\frac{1}{3}\right) e^{-3x} + C$$

⑤ Isolate y

$$y = -2 + C e^{3x}$$

Do Home!

Solve $y' + y = f(x)$ if $y(0) = 0$ and $f(x)$ is 1 for x between 0 and 1 (included) and 0 if x greater than 1 (consider continuity in the solution)

$$\frac{dy}{dx} + y = f(x) \quad y(0) = 0, \quad f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \leftarrow \\ 0 & x > 1 \leftarrow y(1^-) = y(1^+) \end{cases}$$

Final answer $y(x) = \begin{cases} 1 - e^{-x} & \leftarrow \text{linear} \\ (e-1)e^{-x} & \leftarrow \text{separable} \end{cases}$

2.4 Exact Equations $f(x,y)$

- An equation $M(x,y)dx + N(x,y)dy$ is called an exact differential in a region R if it corresponds to the differential of some function $f(x,y)$.
- An equation having an exact differential is an **exact equation**:

$M(x,y)dx + N(x,y)dy = 0$
 exact differential

$M = \frac{\partial f}{\partial x}$
 $N = \frac{\partial f}{\partial y}$

Ex: $\frac{1}{3}x^3y^3 = C \Rightarrow$ Differential $\rightarrow d(\frac{1}{3}x^3y^3) = \frac{d(C)}{=0}$
 $df(x,y) = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = x^2y^3dx + x^3y^2dy$
 partial over x ; if not $x \rightarrow$ constant $\Rightarrow \frac{x^2y^3}{M}dx + \frac{x^3y^2}{N}dy = 0$

- Let us call $M(x,y)dx + N(x,y)dy = 0$ an **exact equation** if $M(x,y)dx + N(x,y)dy$ is an exact differential. A necessary and sufficient condition that $M(x,y)dx + N(x,y)dy$ be an **exact differential**, provided that $M(x,y)$ and $N(x,y)$ be continuous and have continuous first partial derivative on R , is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

- To solve an exact equation, we want $f(x,y) = C$

- Lets assume $\frac{\partial f}{\partial x} = M(x,y)$

- $f(x,y) = \int M(x,y)dx + g(y)$

- And $g'(y) = N(x,y) - \frac{\partial}{\partial y} \left[\int M(x,y)dx \right]$

$$\begin{aligned}
 N &= \frac{\partial f}{\partial y} \\
 &= \frac{\partial}{\partial y} \left[\int M dx + g(y) \right]
 \end{aligned}$$

Using exact equation solve $2xy dx + (x^2 - 1) dy = 0$

Examples: ① $M = 2xy \rightarrow \frac{\partial M}{\partial y} = 2x$
 $N = (x^2 - 1) \rightarrow \frac{\partial N}{\partial x} = 2x$ exact

② $M = \frac{\partial f}{\partial x} \rightarrow \partial f = M dx \rightarrow f(x, y) = \int M dx + g(y) = \int 2xy dx + g(y)$

$f(x, y) = 2y \frac{x^2}{2} + g(y) = xy^2 + g(y)$

③ $N = \frac{\partial f}{\partial y} \rightarrow (x^2 - 1) = \frac{\partial}{\partial y} [xy^2 + g(y)] = x^2 + g'(y)$

$g'(y) = -1 \rightarrow \partial g(y) = -dy \rightarrow g(y) = -y + C$

④ $f(x, y) = xy^2 - y + C \Rightarrow f(x, y) = C_1 \Rightarrow xy^2 - y = C_1$

~~Integrating factor~~ $\Rightarrow P(x, y) dx + Q(x, y) dy = 0$
 $\Rightarrow$ Not exact

- Technique to build an exact equation using a multiplying factor
- IT CAN BE VERY UGLY! So be careful

$\Rightarrow \underbrace{\mu(x, y) P(x, y)}_{M(x, y)} dx + \underbrace{\mu(x, y) Q(x, y)}_{N(x, y)} dy = 0$

$\Rightarrow \frac{\partial \mu}{\partial y} P(x, y) + \mu \frac{\partial P(x, y)}{\partial y} = \frac{\partial \mu}{\partial x} Q(x, y) + \mu \frac{\partial Q(x, y)}{\partial x}$

solve to find $\mu \Rightarrow$ Special case: $\mu(x)$ & $\mu(y)$
 see board.

Let μ be a multiplying factor $\rightarrow \mu P dx + Q dy = 0 \rightarrow$ exact

$\frac{\partial}{\partial y} [\mu P] = \frac{\partial}{\partial x} [\mu Q] \Rightarrow \frac{\partial \mu}{\partial y} P + \mu \frac{\partial P}{\partial y} = \frac{\partial \mu}{\partial x} Q + \mu \frac{\partial Q}{\partial x} \Rightarrow \left[\frac{\mu(x)}{\mu(y)} \right]$

$\partial f(x) : \mu \frac{\partial P}{\partial y} = \mu \frac{\partial Q}{\partial x} + \frac{\partial \mu}{\partial x} Q \Rightarrow \mu(x) = e^{\left(\frac{\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right)}{Q} \right) dx}$

$\partial f(y) \Rightarrow \mu(y) = e^{\left(\frac{\left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y} \right)}{P} \right) dy}$

Ex: $\sin(x) \cos(\ln(y)) dx - \cos(x) \sin(\ln(y)) dy = 0, y(0) = 3$

$\cos(x) \cos(\ln(y)) = \cos(\ln(3))$
 $\begin{cases} \frac{d}{dx} \cos(\ln(x)) = \sin(\ln(x)) \\ \frac{d}{dy} \sin(\ln(y)) = \cos(\ln(y)) \end{cases}$

Ex: $\sin(x) \cos(\ln(y)) dx - \cos(x) \sin(\ln(y)) dy = 0$, $y(0) = 3$

$$\cos(x) \cos(\ln(y)) = \cos(\ln(3)) \quad \begin{cases} \frac{d}{dx} \cos(\ln(x)) = \sin(\ln(x)) \\ \frac{d}{dy} \sin(\ln(y)) = \cos(\ln(y)) \end{cases}$$

Ex: Make $-y dx + x dy^2 = 0$ exact and Solve using exact equation tech.

$$\textcircled{1} \begin{matrix} P(x,y) = -y \rightarrow \frac{\partial P}{\partial y} = -1 \\ Q(x,y) = x \rightarrow \frac{\partial Q}{\partial x} = 1 \end{matrix} \quad \text{Not exact} \quad \textcircled{2} \text{ Integrant for } P: \frac{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}{Q} (M(x))$$

$$\rightarrow \frac{-1-1}{x} = \frac{-2}{x} \Rightarrow M(x) = e^{\int -\frac{2}{x} dx} = e^{-2 \ln|x|} = e^{\ln \frac{1}{x^2}} = \frac{1}{x^2}$$

$$\textcircled{3} \frac{1}{x^2} [-y dx + x dy = 0] \Rightarrow \left[\underbrace{\frac{-y}{x^2}}_M dx + \underbrace{\frac{1}{x}}_N dy = 0 \right] \quad \begin{matrix} \frac{\partial M}{\partial y} = -\frac{1}{x^2} \\ \frac{\partial N}{\partial x} = -\frac{1}{x^2} \end{matrix} \quad \text{exact}$$

$$\textcircled{4} \frac{-y}{x^2} = \frac{\partial f}{\partial x} \Rightarrow \partial f = \frac{-y}{x^2} dx \rightarrow f(x,y) = \int -\frac{y}{x^2} dx + g(y)$$

$$f(x,y) = \frac{y}{x} + g(y)$$

Ex: Make $-y dx + x dy$ exact and Solve using exact equation tech.

$$\textcircled{1} \begin{aligned} P(x,y) &= -y \rightarrow \frac{\partial P}{\partial y} = -1 \\ Q(x,y) &= x \rightarrow \frac{\partial Q}{\partial x} = 1 \end{aligned} \quad \left. \vphantom{\begin{aligned} P(x,y) &= -y \\ Q(x,y) &= x \end{aligned}} \right] \text{exact} \quad \textcircled{2} \text{Integrand for } \mu: \frac{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}{Q} (M(x))$$

$$\begin{aligned} \mu &= \frac{-1 - 1}{x} = -\frac{2}{x} \\ \mu(x) &= e^{\int -\frac{2}{x} dx} = e^{-2 \ln x} = x^{-2} \end{aligned}$$

$$-y dx + x dy = 0 \quad \Rightarrow$$

2.2

In Problems 1-22, solve the given differential equation by separation of variables.

1. $\frac{dy}{dx} = \sin 5x$

Q1: $\frac{dy}{dx} = \sin 5x \Rightarrow dy = \sin 5x dx \Rightarrow \int dy = \int \sin 5x dx$

$\Rightarrow u = 5x \quad du = 5 dx \quad dx = \frac{du}{5}$

① $\Rightarrow \int dy \Rightarrow \int y$

② $\Rightarrow \int \frac{\sin u}{5} du \Rightarrow \frac{1}{5} \int \sin u du \Rightarrow -\frac{1}{5} \cos u \Rightarrow -\frac{1}{5} \cos 5x + C$

Sol: $y = -\frac{1}{5} \cos 5x + C$

7. $\frac{dy}{dx} = e^{3x+2y}$

Q7: $\frac{dy}{dx} = e^{3x+2y} \Rightarrow \frac{dy}{dx} = e^{3x} \cdot e^{2y} \Rightarrow e^{-2y} dy = e^{3x} dx \Rightarrow \int e^{-2y} dy = \int e^{3x} dx$

① $\Rightarrow \int e^{-2y} dy \Rightarrow -\frac{1}{2} e^{-2y}$

② $\Rightarrow \int e^{3x} dx \Rightarrow \frac{1}{3} e^{3x} + C$

$-\frac{1}{2} e^{-2y} = \frac{1}{3} e^{3x} + C \Rightarrow e^{-2y} = -\frac{2}{3} e^{3x} - 2C \Rightarrow -2y = \ln \left(-\frac{2}{3} e^{3x} - 2C \right)$

$\Rightarrow y = -\frac{1}{2} \ln \left(-\frac{2}{3} e^{3x} - 2C \right)$

9. $y \ln x \frac{dx}{dy} = \left(\frac{y+1}{x} \right)^2$

Q9: $y \ln x \frac{dx}{dy} = \left(\frac{y+1}{x} \right)^2 \Rightarrow y \ln x \frac{dx}{dy} = \frac{(y+1)^2}{x^2} \Rightarrow x^2 y \ln x \frac{dx}{dy} = (y+1)^2$

$\Rightarrow x^2 y \ln x dx = (y+1)^2 dy \Rightarrow x^2 \ln x dx = \frac{(y+1)^2}{y} dy$

① $\int x^2 \ln x dx \Rightarrow \begin{matrix} u = \ln x \\ du = \frac{1}{x} dx \\ v = \frac{x^3}{3} \end{matrix}$ Integration by parts ($\int u dv = uv - \int v du$): $= \left(\ln x \right) \left(\frac{x^3}{3} \right) - \int \frac{x^3}{3} \cdot \frac{1}{x} dx = \left(\ln x \right) \left(\frac{x^3}{3} \right) - \frac{1}{3} \int x^2 dx \Rightarrow \ln x \frac{x^3}{3} - \frac{1}{3} \cdot \frac{x^3}{3} \Rightarrow \ln x \frac{x^3}{3} - \frac{x^3}{9}$

② $\int \frac{(y+1)^2}{y} dy \Rightarrow \int \frac{y^2 + 2y + 1}{y} dy \Rightarrow \int \frac{y^2}{y} + \frac{2y}{y} + \frac{1}{y} dy \Rightarrow \int \left(y + 2 + \frac{1}{y} \right) dy \Rightarrow \frac{y^2}{2} + 2y + \ln |y| + C$

Sol: $\ln x \frac{x^3}{3} - \frac{x^3}{9} = \frac{y^2}{2} + 2y + \ln |y| + C$

13. $(e^x + 1)^2 e^{-x} dx + (e^x + 1)^3 e^{-x} dy = 0$

Q13: $(e^x + 1)^2 e^{-x} dx = -(e^x + 1)^3 e^{-x} dy$ divide both sides by $(e^x + 1)^3 (e^x + 1)^2$ $\Rightarrow \frac{e^{-x} dx}{(e^x + 1)^3} = -\frac{e^{-x} dy}{(e^x + 1)^2} \Rightarrow \frac{dx}{(e^x + 1)^3 e^{-x}} = -\frac{dy}{(e^x + 1)^2 e^{-x}} \Rightarrow \frac{e^x dx}{(e^x + 1)^3} = -\frac{e^x dy}{(e^x + 1)^2}$

$\frac{e^x dx}{(e^x + 1)^3} \Rightarrow \begin{matrix} u = e^x + 1 \\ du = e^x dx \end{matrix} \Rightarrow \frac{du}{u^3} \Rightarrow \frac{u^{-2}}{-2} \Rightarrow -\frac{1}{2(e^x + 1)^2}$

$-\frac{e^x dy}{(e^x + 1)^2} \Rightarrow \begin{matrix} u = e^x + 1 \\ du = e^x dx \end{matrix} \Rightarrow \frac{dy}{u^2} \Rightarrow -\frac{u^{-1}}{-1} \Rightarrow \frac{1}{u} \Rightarrow \frac{1}{e^x + 1} + C$

Sol: $-\frac{1}{2(e^x + 1)^2} = \frac{1}{e^x + 1} + C$

19. $\frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$

Q19: $xy + 3x - y - 3 \Rightarrow x(y + 3) - (y + 3) \Rightarrow (x - 1)(y + 3)$
 $xy - 2x + 4y - 8 \Rightarrow x(y - 2) + 4(y - 2) \Rightarrow (x + 4)(y - 2)$

$\frac{dy}{dx} = \frac{(x-1)(y+3)}{(x+4)(y-2)} \Rightarrow \frac{(y-2)}{(y+3)} dy = \frac{(x-1)}{(x+4)} dx$

① $\int \frac{y-2}{y+3} dy \Rightarrow \int 1 - \frac{5}{y+3} dy \Rightarrow y - 5 \ln |y+3| + C$

② $\int \frac{x-1}{x+4} dx \Rightarrow \int 1 - \frac{5}{x+4} dx \Rightarrow x - 5 \ln |x+4|$

Sol: $y - 5 \ln |y+3| + C = x - 5 \ln |x+4|$

In Problems 23–30, find an explicit solution of the given initial-value problem.

25. $x^2 \frac{dy}{dx} = y - xy, y(-1) = -1$ Q25 $x^2 \frac{dy}{dx} = y(1-x) \Rightarrow x^2 \frac{dy}{dx} = y(1-x) \Rightarrow \frac{1}{y} dy = \frac{1}{x^2} - \frac{x}{x^2} dx \Rightarrow \frac{1}{y} dy = x^{-2} - \frac{1}{x} dx$ ① ②

① $\int \frac{1}{y} dy \Rightarrow \ln|y|$ Implicit sol: $\ln|y| = -\frac{1}{x} - \ln|x| + C$

② $\int x^{-2} - \frac{1}{x} dx \Rightarrow -\frac{x^{-1}}{-1} - \ln|x| \Rightarrow -\frac{1}{x} - \ln|x| + C$

Solve for C w $y(-1) = -1 \Rightarrow \ln|-1| = -\frac{1}{-1} - \ln|-1| + C \Rightarrow 0 = 1 - 0 + C \Rightarrow C = -1$

$\ln|y| = -\frac{1}{x} - \ln|x| - 1$

Isolate y $\Rightarrow \ln|y| + \ln|x| = -\frac{1}{x} - 1 \Rightarrow \ln|xy| = -\frac{1}{x} - 1 \Rightarrow |xy| = e^{-\frac{1}{x} - 1} \Rightarrow xy = e^{-\frac{1}{x} - 1}$

$\Rightarrow y = \frac{1}{x} e^{-\frac{1}{x} - 1}$ = Sol explicit

Divide both sides by $\sqrt{1-y^2} \sqrt{1-x^2}$

27. $\sqrt{1-y^2} dx - \sqrt{1-x^2} dy = 0, y(0) = \sqrt{3}/2$ Q27 $\sqrt{1-y^2} dx = \sqrt{1-x^2} dy \Rightarrow \frac{1}{\sqrt{1-x^2}} dx = \frac{1}{\sqrt{1-y^2}} dy$ ① ②

① $\int \frac{1}{\sqrt{1-x^2}} dx \Rightarrow \arcsin x$ Implicit sol: $\arcsin x = \arcsin y + C$

② $\int \frac{1}{\sqrt{1-y^2}} dy \Rightarrow \arcsin y + C$

Solve for C w $y(0) = \frac{\sqrt{3}}{2} \Rightarrow \arcsin(0) = \arcsin\left(\frac{\sqrt{3}}{2}\right) + C \Rightarrow 0 = C + \frac{\pi}{3} \Rightarrow C = -\frac{\pi}{3}$

$\arcsin x = \arcsin y - \frac{\pi}{3} \Rightarrow \arcsin y = \arcsin x + \frac{\pi}{3} \Rightarrow y = \sin\left(\arcsin x + \frac{\pi}{3}\right)$

$\Rightarrow y = \sin\left(\arcsin(x)\right) \cos\left(\frac{\pi}{3}\right) + \cos\left(\arcsin(x)\right) \sin\left(\frac{\pi}{3}\right) \Rightarrow y = \frac{1}{2}x + \sqrt{1-x^2} \cdot \frac{\sqrt{3}}{2}$

Expand using the angle-addition identity for sine ($\sin(A+B) = \sin A \cos B + \cos A \sin B$), where $A = \arcsin(x)$ and $B = \frac{\pi}{3}$:

$y = \sin(\arcsin(x)) \cos\left(\frac{\pi}{3}\right) + \cos(\arcsin(x)) \sin\left(\frac{\pi}{3}\right)$

In Problems 31–34, proceed as in Example 6 and find an explicit solution of the given initial-value problem.

31. $\frac{dy}{dx} = ye^{-x^2}, y(4) = 1 \Rightarrow \frac{1}{y} dy = e^{-x^2} dx$ ① ②

① $\int \frac{1}{y} dy \Rightarrow \ln|y|$

② $\int e^{-x^2} dx \Rightarrow \text{Thanks!}$

2.3

In Problems 1–24, find the general solution of the given differential equation. Give the largest interval over which the general solution is defined. Determine whether there are any transient terms in the general solution.

4. $3 \frac{dy}{dx} + 12y = 4$ Q4 $3 \frac{dy}{dx} + 12y = 4 \Rightarrow \frac{dy}{dx} + 4y = \frac{4}{3} \Rightarrow I(x) = e^{\int 4 dx} = e^{4x} \Rightarrow e^{4x} \frac{dy}{dx} + 4y e^{4x} = \frac{4}{3} e^{4x}$

$\Rightarrow \frac{d(e^{4x}y)}{dx} = \frac{4}{3} e^{4x}$

$\Rightarrow e^{4x}y = \frac{4}{3} \int e^{4x} dx \Rightarrow \frac{4}{3} \cdot \frac{1}{4} e^{4x} = \frac{1}{3} e^{4x} \Rightarrow \frac{1}{3} e^{4x}$

$\Rightarrow e^{4x}y = \frac{1}{3} e^{4x} + C$

$\Rightarrow y = \frac{1}{3} + \frac{C}{e^{4x}}$

$\Rightarrow \frac{1}{3} + \frac{C}{e^{4x}} \Rightarrow \frac{1}{3} + C e^{-4x}$

7. $x^2 y' + xy = 1$ Q7 $x^2 \frac{dy}{dx} + xy = 1$ Divide both sides by x^2

$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2} \Rightarrow \frac{dy}{dx} + \frac{1}{x} y = \frac{1}{x^2} \Rightarrow e^{\int \frac{1}{x} dx} \Rightarrow e^{\ln|x|} = x$

$\Rightarrow \frac{d(xy)}{dx} = \frac{1}{x^2} \Rightarrow \frac{1}{x^2} x dx$

$$dx \cdot \frac{1}{x} = \frac{1}{x^2} \Rightarrow dx \cdot x^{-1} = x^{-2} \Rightarrow e^{\dots} \Rightarrow \dots = \dots$$

$$\Rightarrow \left(\frac{d(xy)}{dx} \right) = \int \frac{1}{x^2} x dx$$

$$\Rightarrow xy = \frac{1}{x}$$

$$\Rightarrow xy = \ln|x| + C$$

$$\Rightarrow y = \frac{\ln|x| + C}{x} \Rightarrow y = \frac{1}{x} \ln|x| + \frac{C}{x}$$

9. $x \frac{dy}{dx} - y = x^2 \sin x$ Q9 $x \frac{dy}{dx} - y = x^2 \sin x$ $\Rightarrow \frac{dy}{dx} - \frac{1}{x} y = \frac{x^2 \sin x}{x} \Rightarrow \frac{dy}{dx} - \frac{1}{x} y = x \sin x$

$$\Rightarrow e^{\int P(x) dx} \Rightarrow e^{\int -\frac{1}{x} dx} \Rightarrow e^{-\ln|x|} = \frac{1}{x}$$

$$\Rightarrow \left(\frac{d(\frac{1}{x} y)}{dx} \right) = \int x \sin x \cdot \frac{1}{x} dx$$

$$\Rightarrow \frac{1}{x} y = \int \sin x dx$$

$$\Rightarrow \frac{1}{x} y = -\cos(x) + C$$

$$\Rightarrow y = x \cdot (-\cos(x)) + C$$

23. $x \frac{dy}{dx} + (3x+1)y = e^{-3x}$ Q23 $x \frac{dy}{dx} + (3x+1)y = e^{-3x} \Rightarrow \frac{dy}{dx} + 3 + \frac{1}{x} = \frac{e^{-3x}}{x}$

$$\Rightarrow e^{\int P(x) dx} \Rightarrow e^{\int 3 + \frac{1}{x} dx} \Rightarrow e^{3x + \ln|x|} \Rightarrow e^{3x} x$$

$$\Rightarrow \frac{d(e^{3x} xy)}{dx} = \frac{e^{-3x}}{x} \cdot e^{3x} xy$$

$$\Rightarrow \int \frac{d(e^{3x} xy)}{dx} = \int \frac{e^{-3x}}{x} \cdot e^{3x} x dx$$

$$\Rightarrow e^{3x} xy = \int 1 dx$$

$$\Rightarrow e^{3x} xy = x + C$$

$$\Rightarrow y = \frac{x+C}{e^{3x} x}$$

27. $L \frac{di}{dt} + Ri = E; i(0) = i_0, L, R, E,$ and i_0 constants Q27

31. $\left(\frac{e^{-2\sqrt{x}} - y}{\sqrt{x}} \right) \frac{dy}{dx} = 1, y(1) = 1$ Q31 $\left(\frac{e^{-2\sqrt{x}} - y}{\sqrt{x}} \right) = \frac{dy}{dx} \Rightarrow \frac{e^{-2\sqrt{x}} - y}{\sqrt{x}} = \frac{dy}{dx} \Rightarrow \frac{dy}{dx} + \frac{1}{\sqrt{x}} y = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$

$$\Rightarrow e^{\int P(x) dx} \Rightarrow e^{\int \frac{1}{\sqrt{x}} dx} \Rightarrow e^{2\sqrt{x}}$$

$$\Rightarrow \left(\frac{d(e^{2\sqrt{x}} y)}{dx} \right) = \int \frac{e^{-2\sqrt{x}}}{\sqrt{x}} \cdot e^{2\sqrt{x}} dx$$

$$\Rightarrow e^{2\sqrt{x}} y = 2x^{1/2} + C$$

W $x=1$ and $y=1 \Rightarrow e^{2\sqrt{1}} 1 = 2 \cdot 1^{1/2} + C \Rightarrow e^2 = 2 + C \Rightarrow C = e^2 - 2$

$$\Rightarrow e^{2\sqrt{x}} y = 2\sqrt{x} + e^2 - 2$$

$$\Rightarrow y = \frac{2\sqrt{x} + e^2 - 2}{e^{2\sqrt{x}}}$$

2.4

In Problems 1–20, determine whether the given differential equation is exact. If it is exact, solve it.

3. $(5x+4y) dx + (4x-8y^2) dy = 0$ Q3

$M(x,y) = (5x+4y)$ ① Check if exact

$N(x,y) = (4x-8y^2)$ $\frac{\partial M}{\partial y} = 4$

$\frac{\partial N}{\partial x} = 4$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

$$\frac{dy}{dx} = 4 \quad \left. \begin{array}{l} \text{or } y \\ \frac{dy}{dx} = 4 \end{array} \right\} = \checkmark$$

② Integrate $M(x, y)$ with respect to x

$$F(x, y) = \int M(x, y) dx = \int (5x + 4y) dx$$

$$F(x, y) = \frac{5x^2}{2} + 4xy + g(y)$$

③ Differentiate with respect to y and match $N(x, y)$

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y} \left(\frac{5x^2}{2} + 4xy + g(y) \right) = 4x + g'(y)$$

match

$$4x + g'(y) = 4x - 8y^3$$

$$g'(y) = -8y^3$$

④ Integrate to find $g(y)$

$$g(y) = - \int 8y^3 dy$$

$$g(y) = - \frac{8y^4}{4} + C \Rightarrow g(y) = -2y^4$$

$$F(x, y) = \frac{5x^2}{2} + 4xy - 2y^4 = C$$

5. $(2xy^2 - 3) dx + (2x^2y + 4) dy = 0$
 $M(x, y)$ $N(x, y)$

① Verify if exact

$$\frac{\partial M}{\partial y} = 4xy$$

$$\frac{\partial N}{\partial x} = 4xy$$

② Integrate $M(x, y)$ with respect to x

$$f(x, y) = \int M(x, y) dx = \int (2xy^2 - 3) dx$$

$$F(x, y) = \frac{2x^2y^2}{2} - 3x + g(y)$$

③ Differentiate with respect to y and match $N(x, y)$

$$\frac{\partial F(x, y)}{\partial y} = \frac{\partial}{\partial y} (x^2y^2 - 3x + g(y)) = 2x^2y + g'(y)$$

match

$$2x^2y + g'(y) = 2x^2y + 4$$

$$g'(y) = 4$$

④ Integrate to find $g(y)$

$$g(y) = \int 4 dy$$

$$g(y) = 4y$$

$$f(x, y) = x^2y^2 - 3x + 4y = C$$

9. $(x - y^3 + y^2 \sin x) dx + (-3xy^2 - 2y \cos x) dy = 0$
 $M(x, y)$ $N(x, y)$

① $\frac{\partial M}{\partial y} = -3y^2 + 2y \sin x$

$\frac{\partial N}{\partial x} = -3y^2 - 2y(-\sin x) = -3y^2 + 2y \sin x$ } exact

② $\int$ respect x

$$F(x, y) = \int M(x, y) dx = \int (x - y^3 + y^2 \sin x) dx$$

$$\Rightarrow \frac{x^2}{2} - xy^3 - y^2 \cos x + g(y)$$

③ Diff. respect y

$$\frac{\partial F(x,y)}{\partial y} = \frac{\partial}{\partial y} \left(\frac{x^2}{2} - xy^3 - y^2 \cos x + g(y) \right)$$

$$\Rightarrow -3xy^2 - 2y \cos x + g'(y)$$

Match

$$-3xy^2 - 2y \cos x + g'(y) = -3xy^2 - 2y \cos x$$

$$g'(y) = 0$$

④

$$g(y) = \int 0 dy$$

$$\frac{x^2}{2} - xy^3 - y^2 \cos x = 0$$

15. $\left(x^2y^3 - \frac{1}{1+y^2}\right) \frac{dx}{dy} + x^3y^2 = 0$ Q15 $\left(x^2y^3 - \frac{1}{1+y^2}\right) dx + x^3y^2 dy = 0$

$M(x,y)$

$N(x,y)$

$$\textcircled{1} \frac{\partial M}{\partial y} = 3x^2y^2 \quad \frac{\partial N}{\partial x} = 3x^2y^2 \quad \left\{ \text{exact} \right.$$

$$\textcircled{2} F(x,y) = \int M(x,y) dx = \int x^2y^3 - \frac{1}{1+y^2} dx$$

$$f(x,y) = \frac{x^3y^3}{3} - \frac{1}{3} \arctan(y) + g(y)$$

③

$$\frac{\partial f(x,y)}{\partial y} = \frac{\partial}{\partial y} \left(\frac{x^3y^3}{3} - \frac{1}{3} \arctan(y) + g(y) \right)$$

$$\frac{x^3}{3} \cdot 3 \cdot \frac{y^2}{3} + g'(y) = x^3y^2 + g'(y)$$

match

$$x^3y^2 + g'(y) = x^3y^2$$

$$\textcircled{4} g'(y) = 0$$

$$\frac{x^3y^3}{3} - \frac{1}{3} \arctan(y) = C$$

27. $(y^3 + kxy^4 - 2x) dx + (3xy^2 + 20x^2y^3) dy = 0$ Q17 $\frac{\partial M}{\partial y} = 3y^2 + 4kxy^3 \quad \left\{ k=10 \text{ exact} \right.$

$M(x,y)$

$N(x,y)$

①

$$\frac{\partial N}{\partial x} = 3y^2 + 40xy^3$$

$$\textcircled{2} \int \text{with } x$$

$$f(x,y) = \int (y^3 + 10xy^4 - 2x) dx$$

$$f(x,y) = xy^3 + 5x^2y^4 - x^2 + g(y)$$

③

$$\frac{\partial f(x,y)}{\partial y} = \frac{\partial}{\partial y} (xy^3 + 5x^2y^4 - x^2 + g(y))$$

$$3xy^2 + 20x^2y^3 + g'(y)$$

Match

$$3xy^2 + 20x^2y^3 + g'(y) = 3xy^2 + 20x^2y^3$$

$$g'(y) = 0$$

④

$$xy^3 + 5x^2y^4 - x^2 = C$$

29. $(-xy \sin x + 2y \cos x) dx + 2x \cos x dy = 0; \quad \mu(x,y) = xy$

$M(x,y)$

$N(x,y)$

$\mu(x,y)$

$$xy + 5x y' - x^2 = 0$$

29. $(-xy \sin x + 2y \cos x) dx + 2x \cos x dy = 0$; $\mu(x, y) = xy$

$$M(x, y) \quad N(x, y) \quad \textcircled{0} \quad \frac{dM}{dy} =$$

Quiz 1 prep

September 21, 2026 12:27 PM

Quiz 1 Mushfiqur

$$yy' + x^2 = 4 \quad y(0) = 2 \Rightarrow y \frac{dy}{dx} + x^2 = 4$$

$$\Rightarrow y dy = -x^2 + 4 dx$$

$$\textcircled{1} \int y dy = \frac{y^2}{2}$$

$$\textcircled{2} \int -x^2 + 4 dx = -\frac{x^3}{3} + 4x + C$$

$$\Rightarrow \frac{y^2}{2} = -\frac{x^3}{3} + 4x + C \quad \begin{matrix} x=0 \\ y=2 \end{matrix} \Rightarrow \frac{2^2}{2} = 0 + 0 + C \Rightarrow C = 2$$

$$\Rightarrow \frac{y^2}{2} = -\frac{x^3}{3} + 4x + 2 \Rightarrow y^2 = -\frac{2x^3}{3} + 8x + 4 \Rightarrow y = \sqrt{-\frac{2x^3}{3} + 8x + 4}$$

$$yy' + x^2 = 4 \quad y(0) = 2$$

$$y \frac{dy}{dx} + x^2 = 4 \Rightarrow y dy = 4 - x^2 dx$$

$$\textcircled{1} \int y dy = \frac{y^2}{2}$$

$$\textcircled{2} \int 4 - x^2 dx = 4x - \frac{x^3}{3} + C$$

$$\left. \begin{matrix} \textcircled{1} \int y dy = \frac{y^2}{2} \\ \textcircled{2} \int 4 - x^2 dx = 4x - \frac{x^3}{3} + C \end{matrix} \right\} \frac{y^2}{2} = 4x - \frac{x^3}{3} + C \quad \begin{matrix} x=0 \\ y=2 \end{matrix} \quad \frac{2^2}{2} = 0 - 0 + C \Rightarrow 2 = C$$

$$\frac{y^2}{2} = 4x - \frac{x^3}{3} + 2 \Rightarrow y^2 = 8x - \frac{2x^3}{3} + 4 \Rightarrow y = \sqrt{8x - \frac{2x^3}{3} + 4}$$

$$4. \quad 3 \frac{dy}{dx} + 12y = 4 \quad \Rightarrow \quad \frac{dy}{dx} + 4y = \frac{4}{3} \quad \Rightarrow \quad e^{\int P(x) dx} \Rightarrow e^{\int 4 dx} \Rightarrow e^{4x}$$

$$\Rightarrow \left(\frac{d(e^{4x}y)}{dx} \right) = e^{4x} \cdot \frac{4}{3}$$

$$\Rightarrow e^{4x} y = \frac{1}{4} \cdot \frac{4}{3} \cdot e^{4x} + C$$

$$\Rightarrow e^{4x} y = \frac{1}{3} e^{4x} + C \Rightarrow y = \frac{\frac{1}{3} e^{4x} + C}{e^{4x}} \Rightarrow y = \frac{1}{3} + \frac{C}{e^{4x}}$$



Lecture 3

ENGR 213

2.5 Substitutions

- The goal is to reduce the form of the equation to a simpler form.
- Very useful in engineering: fluid mechanics, advance thermodynamics, heat transfer, dynamics, etc.
- 3 techniques shown in this course:
 - Reduction to separable form
 - Bernoulli equation
 - Linear substitution

Reduction to separable form:

- Condition of use: equation $R(x,y)dx + S(x,y)dy = 0$ MUST be "homogeneous"
- $F(x,y)$ is "homogeneous" if $f(tx,ty) = t^k f(x,y)$
- Idea: substitute one variable by a ratio, for example $u = y/x$
- Hence we can have $R(x,y) = x^k R(1,u)$ and $S(x,y) = x^k S(1,u)$
 $u = \frac{y}{x}$ or $v = \frac{x}{y}$
- We try to find a common factor to both R and S
- Let's see an example

Example: Solve $(x^2 + y^2) dx + (x^2 - xy) dy = 0$

① $(x^2 + y^2)$
 $x^2(1 + \frac{y^2}{x^2})$

$(x^2 - xy)$
 $x^2(1 - \frac{y}{x})$

② Substitution: $u = \frac{y}{x}$
 $y = ux \rightarrow dy = u dx + x du$

③ re-write with $u/x \Rightarrow x^2(1+u^2) dx + x^2(1-u)(u dx + x du) = 0$

$$x^2 dx + x^2 u^2 dx + x^2 u dx + x^3 du - x^2 u^2 dx - x^2 u du = 0$$

$$x^2(1+u) dx + x^3(1-u) du = 0$$

④ Separate & Integrate $\rightarrow x^2(1+u) dx = x^3(u-1) du$

$$\frac{x^2}{x^3} dx = \frac{(u-1)}{(u+1)} du \leftarrow$$

$$\frac{x^2}{x^3} dx = \frac{(u-1)}{(u+1)} du \rightarrow \frac{dx}{x} = \frac{u(1-1)-1}{(u+1)} du = \left[1 - \frac{2}{u+1}\right] du$$

$$\text{Integrate} \rightarrow \int \frac{dx}{x} = \int \left[1 - \frac{2}{(u+1)}\right] du$$

$$C + \ln|x| = u - 2 \ln|u+1|$$

⑤ Go back to x & $y \rightarrow \ln|x| + C = \frac{y}{x} - 2 \ln\left|\frac{y}{x} + 1\right| \leftarrow \text{this is fine}$

extra: $\ln|x| + C = \frac{y}{x} + \ln \frac{1}{\left[\frac{y}{x} + 1\right]^2}$

Bernoulli equation

- Named after Jacques Bernoulli (circa 1695) for his use of the technique to solve the isochrone problem

- Equation of the form:

$$\frac{dy}{dx} + P(x)y = f(x)y^n$$

- Substitution: $u = y^{1-n}$, $n \neq 0$, $n \neq 1$
- Reduction to a linear form!
- $u' = (1-n) y^{-n} y'$ and $1/(1-n) u' + P(x) u = f(x) \dots$
- example

$$\textcircled{3} \frac{du}{dx} + (1-n)P(x)u = (1-n)f(x)$$

$$\textcircled{1} u = \dots$$

$$\textcircled{2} u = y$$

Ex: Use Bernoulli sub to solve $x \frac{dy}{dx} + y = x^2 y^2$

$\Rightarrow$ Standard form $\frac{dy}{dx} + \frac{1}{x}y = x y^2$

$$\textcircled{1} n = 2$$

$$\textcircled{2} u = y^{1-n} = y^{1-2} = \frac{1}{y}$$

$$(1-n) = -1$$

$$\textcircled{3} \frac{du}{dx} + (-1) \frac{1}{x} u = (-1) x$$

$$\frac{du}{dx} - \frac{1}{x} u = -x \rightarrow e^{\int P(x) dx} = e^{\int -\frac{1}{x} dx} = e^{-\ln|x|} = \frac{1}{x}$$

$$d[e^{\int P(x) dx} u] = e^{\int P(x) dx} f(x) dx$$

$$\Rightarrow d\left[\frac{u}{x}\right] = \frac{1}{x} \cdot (-x) dx \rightarrow \frac{u}{x} = -x + C \rightarrow u = Cx - x^2$$

$$\frac{1}{y} = Cx - x^2 \rightarrow y(x) = \frac{1}{Cx - x^2}$$

Other type

$$\text{ex: } \sin(3x+4y) = \frac{dy}{dx}$$

$$(3x+8y)^2 + \frac{dy}{dx} = 0$$

- Another type of substitution you might see is :

$$u = Ax + By + C, B \neq 0$$

- Generally more complex, no general formulas.

- Example:

$$\text{Solve } \frac{dy}{dx} = (-2x + y)^2 - 7, \text{ for } y(0)=0$$

$$u = -2x + y \leadsto \frac{dy}{dx} = u^2 - 7 \leadsto \frac{du}{dx} = -2 + \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{du}{dx} + 2$$

$$\frac{du}{dx} + 2 = u^2 - 7 \leadsto \frac{du}{dx} = u^2 - 9 \leadsto \frac{du}{(u^2 - 9)} = dx$$

$$\frac{1}{(u^2 - 9)} \text{ using partial fraction}$$

$$\frac{1}{(u-3)(u+3)}$$

$$\frac{1}{(u-3)(u+3)} = \frac{A}{(u-3)} + \frac{B}{(u+3)} \Rightarrow \frac{1}{(u^2 - 9)} = \frac{A(u+3) + B(u-3)}{(u^2 - 9)}$$

$$u=3 \rightarrow 1=6A \Rightarrow A=\frac{1}{6}$$

$$u=-3 \rightarrow B=-\frac{1}{6}$$

$$\int \frac{1}{6} \left[\frac{1}{(u-3)} - \frac{1}{(u+3)} \right] du = \int dx \Rightarrow \frac{1}{6} [\ln|u-3| - \ln|u+3|] = x + C$$

$$\frac{1}{6} \ln \frac{|u-3|}{|u+3|} = x + C$$

isolate u

$$u = \frac{3(1 + C_1 e^{6x})}{3(1 - C_1 e^{6x})}$$

knowing

$$u = -2x + y \leadsto -2x + y = \frac{3(1 + C_1 e^{6x})}{3(1 - C_1 e^{6x})}$$

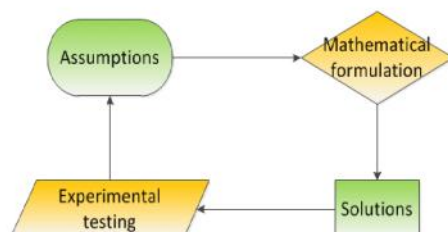
$$\Rightarrow y = 2x + 3 \frac{[1 - e^{6x}]}{[1 + e^{6x}]}$$

1.3/2.7: Linear Model.

- Remember the goal of ODE:

$$\text{proportional} \leadsto k$$

$$\text{rate of change} \leadsto \frac{d}{dt}$$



- Model the real world.

Growth/Decay Model:

- ~~Suppose that an archeologist excavates a bone and measures its content of carbone 14.~~ ^{Not important} If the content is ^{30%} the value of a living bone, find the approximate age of the bone.

$$\frac{dx}{dt} = kx, \quad x(t_0) = x_0$$

$x_0 \rightarrow$ living bone
 $t_0 = 0$
 t for $x = 0,3 x_0$
 $\text{if } t = 5730$
 $x = \frac{1}{2} x_0$

Use a half life of 5730 years

① $x(t)$ ② t for $x = 0,3 x_0$

$$\frac{dx}{dt} = kx \rightarrow \frac{dx}{x} = k dt \rightarrow x(t) = C e^{kt} \quad \begin{matrix} C=? \\ k=? \end{matrix}$$

$$\text{IVP: } x(0) = x_0 = C \rightarrow x(t) = x_0 e^{kt}$$

$$x(5730) = \frac{1}{2} x_0 = x_0 e^{k \cdot 5730} \rightarrow \ln \frac{1}{2} = k \cdot 5730 \rightarrow k = \frac{\ln(1/2)}{5730} \approx -0,00012096$$

$$x(t) = x_0 e^{-0,00012096t}$$

$$\Rightarrow \text{for } x = 0,3 x_0 \rightarrow 0,3 x_0 = x_0 e^{-0,00012096t} \rightarrow \ln(0,3) = -0,00012096t \rightarrow t = 11460 \text{ years}$$

Newton's law of cooling

- You turn off the heat 2 hours before going to bed (time 0). The temperature is 20°C. When you go to bed, the temperature has dropped to 18°C. If the exterior temperature is 4°C, what will be the temperature in your room after 8 hours of sleep?

$$\begin{aligned} T &\rightarrow \text{studied object} \\ T_{ext} &= 4 & T(0) &= ? \\ T(0) &= 20^\circ\text{C} & T &= 12,207^\circ\text{C} \\ T(2) &= 18^\circ\text{C} \end{aligned}$$

$$\frac{dT}{dt} = k(T - T_{ext})$$

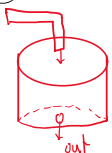
Mixtures $\frac{dx}{dt} = \text{In} - \text{Out}$

$x = \text{mass of chemical}$

Assume a tank having a constant volume of liquid at 300 gal. Initially 50 lbs of salt are dissolved in it. Consider input of brine at 3 gal/min at a concentration of $\frac{2 \text{ lb}}{\text{gal}}$ of salt.

(a) Find amount of salt as a function of time

(b) Max amount of salt



$V = 300 \text{ gal}$

$x \rightarrow \text{salt}$

$x(0) = 50 \text{ lbs}$

Input: $\frac{3 \text{ gal}}{\text{min}} \cdot \frac{2 \text{ lb}}{\text{gal}} \Rightarrow \frac{6 \text{ lbs}}{\text{min}}$

Output: $\frac{3 \text{ gal}}{\text{min}} \cdot \frac{x}{300 \text{ gal}} = \frac{x}{100} \frac{\text{lbs}}{\text{min}}$

$$\frac{dx}{dt} = 6 - \frac{x}{100} \Rightarrow \frac{dx}{dt} = \frac{600 - x}{100}$$

$$\int \frac{dx}{(600-x)} = \int \frac{dt}{100} \Rightarrow -\ln|600-x| = \frac{t}{100} + C$$

$$\ln|600-x| = \frac{-t}{100} - C \xrightarrow{\text{exp.}} 600-x = C_1 e^{-t/100}$$

$$x(t) = 600 - C_1 e^{-t/100} \rightarrow x(0) = 50 = 600 - C_1 \rightarrow C_1 = 550$$

$$x(t) = 600 - 550 e^{-t/100}$$

$$\textcircled{b} \lim_{t \rightarrow \infty} x(t) = 600$$

Circuit: brief info

- 2 cases for this chapter: $i = \text{current}$
 - Current: LR circuit (L is inductor, henry ; R is resistor, ohm)

$$L \frac{di}{dt} + Ri = E(t)$$

- Charge: RC circuit (C is capacitor, farad) $q = \text{charge}$

$$R \frac{dq}{dt} + \frac{q}{C} = E(t)$$

2.8: Non-linear Model

- Logistic Equation
- 1838 by Pierre-François Verhulst

cur

$$Ax + By + C, B \neq 0$$

no general formulas.

$$\frac{dy}{dx} = A + \frac{B}{C} \frac{1}{x}$$

$$= u^2$$

$$\int \frac{du}{1+u^2} = \int dx$$

$$x + C$$

$$y = x + C$$

$$(x+C)$$

$$-x-1$$

$$\frac{dy}{dx}$$

$$\Rightarrow \frac{du}{dx} = \sec^2 u \rightarrow \frac{1}{\cos^2 u}$$

$$x^2(1+\sqrt{u}) du + x u^{3/2} dx = 0 \quad \div x$$

$$x(1+\sqrt{u}) du + u^{3/2} dx = 0$$

$$u^{3/2} dx = -x(1+\sqrt{u}) du$$

$$\textcircled{1} \frac{dx}{x} = \frac{1+\sqrt{u}}{u^{3/2}} du$$

$$\textcircled{2} \int \frac{dx}{x} = \ln|x| + C$$

$$\textcircled{2} \int \frac{1}{u^{3/2}} + \left(\frac{\sqrt{u}}{u^{3/2}} du \right) = \int \frac{1}{u^{3/2}} + \int \frac{1}{u} = -\frac{1}{\frac{1}{2}} + \ln|u| = -2 \frac{1}{u^{1/2}} + \ln|u|$$

$$-2 u^{-1/2} + \ln|u| = \ln|x| + C \approx -2 \left(\frac{y}{x} \right)^{-1/2} + \ln \left| \frac{y}{x} \right| = \ln|x| + C$$

$$\Rightarrow C = -2 \left(\frac{y}{x} \right)^{-1/2} + \ln \left| \frac{y}{x} \right| - \ln|x|$$

$$= C = -2 \sqrt{\frac{x}{y}} + \ln|y|$$

In Problems 11–14, solve the given initial-value problem.

13. $(x + y e^{xy}) dx - x e^{xy} dy = 0, y(1) = 0$

19. $x^2 \frac{dy}{dx} + y^2 = \frac{1}{x} y \Rightarrow \frac{dy}{dt} = \frac{y}{x} - \frac{y^2}{t^2} \quad u = \frac{y}{x} \quad y = ut$

$$\frac{dy}{dt} = u - u^2$$

$$\frac{dy}{dt} = ut \times \frac{du}{dx}$$

$$ut \times \frac{du}{dx} = u - u^2$$

$$x \frac{du}{dx} = -u^2 \Rightarrow \left(\frac{du}{-u^2} \right) = \left(\frac{dx}{x} \right) \textcircled{1}$$

$$\textcircled{1} \int \frac{du}{-u^2} = \frac{1}{u}$$

$$\textcircled{2} \int \frac{dx}{x} = \ln|x| + C$$

$$\frac{1}{u} = \ln|x| + C$$

$$\frac{t}{y} = \ln|x| + C$$

† #3 2.7 linear models

#6

$$A(0) = 100 \text{ mg}$$

After 6h, the mass
has decreased

by 3% $\rightarrow$ so 97%

remains

$$A(6) = 97$$

Because the rate of
decay is proportional
to the amount present

$$\frac{dA}{dt} = kA$$

$$\rightarrow \int \frac{dA}{A} = \int k dt$$

$$\rightarrow \ln|A| = kt + C \rightarrow A(t) = A_0 e^{kt}$$

$$A(0) = 100 \rightarrow A(t) = 100 e^{kt}$$

$$A(6) = 97 = 100 e^{6k} = 0.97 \rightarrow k = \frac{1}{6} \ln(0.97)$$

$$A(24) = 100 e^{24k} = 100 (e^{6k})^4 = 100 (0.97)^4$$

$$\rightarrow A(24) = 88.53 \text{ mg}$$

#21

$$t=0, T=98.6^{\circ}\text{F}$$

$$t=t_d, T=85^{\circ}\text{F}$$

$$t=t_d+1, T=80^{\circ}\text{F}$$

$$t_d=?$$

Write Newton's cooling equation

$$\frac{dT}{dt} = k(T - T_s)$$

$s = \text{surrounding}$

$$\rightarrow \frac{dT}{T - T_s} = k dt \rightarrow \ln|T - T_s| = kt + C_0$$

$$\rightarrow T(t) - T_s = C e^{kt}$$

$$T(0) = 98.6 \rightarrow 98.6 - 70 = C \rightarrow C = 28.6$$

$$T(t) = 70 + 28.6 e^{kt}$$

$$T(t_d) = 85 \rightarrow 15 = 28.6 e^{kt_d}$$

$$T(t_d+1) = 80 \rightarrow 10 = 28.6 e^{k(t_d+1)} \quad \left. \vphantom{\begin{matrix} T(t_d) = 85 \\ T(t_d+1) = 80 \end{matrix}} \right\} \frac{10}{15} = \frac{28.6 e^{k(t_d+1)}}{28.6 e^{kt_d}} = e^k$$

$$\rightarrow e^k = \frac{2}{3} \rightarrow k = \ln \frac{2}{3}$$

$$15 = 28.6 e^{kt_d} \rightarrow e^{kt_d} = \frac{15}{28.6} \rightarrow kt_d = \ln\left(\frac{15}{28.6}\right)$$

$$\rightarrow t_d = \frac{\ln\left(\frac{15}{28.6}\right)}{\ln\left(\frac{2}{3}\right)} = 1.59 \text{ h} \rightarrow t_d = 1 \text{ h } 36 \text{ min}$$



Lecture 4

ENGR 213

2.8: Non-linear Model

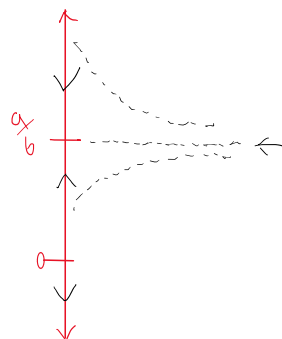
logistic equation
chemical reaction
leaking tank

- Logistic Equation
- 1838 by Pierre-François Verhulst

$$\frac{dP}{dt} = P(a - bP) \quad a > 0 \quad (a > b)$$

$$\frac{dP}{dt} = aP - bP^2 \quad aP \rightarrow \text{growth}$$

$$\quad \quad \quad -bP^2 \rightarrow \text{stop growth}$$



example

$x \rightarrow \#$ of students (infected)

$$\frac{dx}{dt} = kx(1000-x) \rightarrow \frac{dx}{dt} = 1000kx - kx^2$$

$$\begin{aligned} x(0) &= 1 \\ x(4) &= 50 \\ x(6) &= ? \end{aligned}$$

- A student with flu virus return to an isolated college of 1000 students. If the rate of spreading of the disease is proportional to the infected students x and to the non-infected ones, find the number of infected student after 6 days if 50 are infected after 4 days.

bernoulli: $\rightarrow \frac{dx}{dt} - 1000kx = -kx^2$

$$\begin{aligned} n &= 2 \\ u &= x^{-1} = \frac{1}{x} \quad 1-n = -1 \end{aligned}$$

$$\boxed{\frac{du}{dt} + 1000ku = k}$$

linear $\rightarrow e^{\int 1000k dt} = e^{1000kt} = d[e^{1000kt} u] = ke^{1000kt} dt$

$$e^{1000kt} u = \frac{k}{1000k} e^{1000kt} + C \rightarrow u = \frac{1}{e^{1000kt}} \left[\frac{K e^{1000kt}}{1000k} + 1000kC \right]$$

$$u = \frac{k + 1000kC e^{-1000kt}}{1000k} \rightarrow x = \frac{1000}{1 + 1000C e^{-1000t}}$$

ICP $\rightarrow x(0) = 1 = \frac{1000}{1 + 1000C} \rightarrow 1 + 1000C = 1000 \rightarrow C = \frac{999}{1000}$

$$e^{1000kt} u = \frac{k}{1000k} e^{1000kt} + C \rightarrow u = \frac{1}{e^{1000kt}} \left[\frac{k e^{1000kt}}{1000k} + 1000kC \right]$$

$$u = \frac{k + 1000kC e^{-1000kt}}{1000k} \rightarrow x = \frac{1000}{1 + 1000C e^{-1000kt}}$$

$$\text{IUP} \rightarrow x(0) = 1 = \frac{1000}{1 + 1000C} \rightarrow 1 + 1000C = 1000 \rightarrow C = \frac{999}{1000}$$

$$x(t) = \frac{1000}{1 + 999 e^{-1000kt}} \rightarrow x(4) = 50 \rightarrow -1000k = -0.49906$$

$$x(t) = \frac{1000}{1 + 999 e^{-0.49906t}} \quad x(6) = 276.247 \rightarrow 276 \text{ students}$$

↓

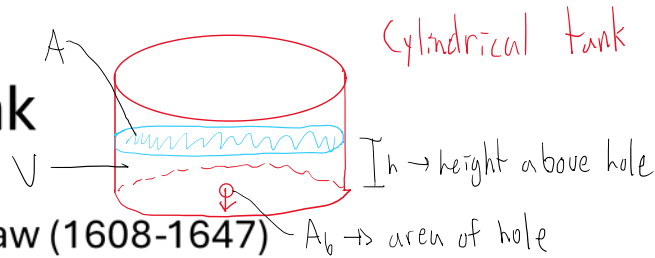
Other models: $A \& B \rightarrow C$ $x \rightarrow \text{amount of } C$
 $\left. \begin{array}{l} \text{ag of } A \\ \text{bg of } B \end{array} \right\} \text{initially}$ $M \rightarrow \text{part of } A \text{ in } C$
 $N \rightarrow \text{part of } B \text{ in } C$

- The next few models are less likely to be used, but they present some nice features to understand.

- 1- Chemical Reactions (explanation only)

$$\frac{dx}{dt} = k \left[\frac{a(M+N)}{M} - x \right] \left[b \frac{(M+N)}{N} - x \right] \rightarrow \text{Partial fraction see textbook}$$

2- Leaking tank



- Use of Torricelli's law (1608-1647)

Velocity of fluid $v = \sqrt{2gh}$ $g = 9.81 \text{ m/s}^2$

Cylindrical tank → $V = Ah$

$$\Rightarrow \frac{dV}{dt} = \text{velocity}(v) \cdot A_b \rightarrow \frac{dV}{dt} = A_b \sqrt{2gh}$$

$$A \frac{dh}{dt} = A_b \sqrt{2gh} \rightarrow \frac{dh}{dt} = \frac{A_b}{A} \sqrt{2gh}$$

* For conical tank:  $V = \frac{1}{3} Ah$ $V = \frac{1}{3} \pi \sin^2 \theta h^3$

Midterm #1 oct 19

Chap 2 + 17.1+17.2

Chapter 17.1 : Intro to Complex Numbers

$\mathbb{C} \rightarrow$ complex set

- The goal is to provide a quick introduction to Complex numbers
- A complex number, z , is a number in the complex set, represented by the sum of two information:

$z = x + iy$, where x is a real number, y an imaginary ~~number~~ part

The number $i = \sqrt{-1}$ is called the imaginary number

$$\boxed{\text{Re}(z) = x}$$

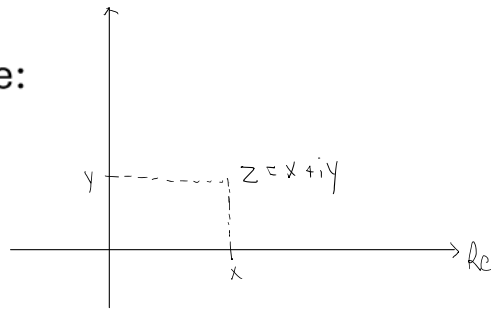
Real part

and

$$\boxed{\text{Im}(z) = y}$$

Imaginary part

- **Complex plane:**
(Argand diagram)



$$z_1 = x_1 + iy_1$$

$$z_2 = x_2 + iy_2$$

- **Basics Properties:**

- Two complex numbers are equal if and only if both the real and the imaginary part are equal

$$z_1 = z_2 \Rightarrow \begin{matrix} x_1 = x_2 \\ y_1 = y_2 \end{matrix}$$

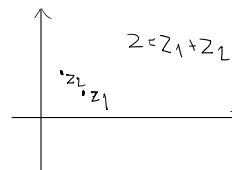
- The complex conjugate, $\bar{z}$, of the complex number $z = x + iy$ is defined as:

$$\bar{z} = x - iy$$

$$z = 3 + 4i: \begin{matrix} \text{Re} = 3 \\ \text{Im} = 4 \end{matrix}$$

- The sum/difference of complex numbers are done by adding/subtracting together the real parts, and together the imaginary parts:

$$\begin{matrix} z_1 = x_1 + iy_1 \\ z_2 = x_2 + iy_2 \end{matrix} \left\{ \begin{matrix} z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2) \end{matrix} \right.$$



$$\begin{matrix} z = x + iy \\ \bar{z} = x - iy \end{matrix}$$

$$\begin{matrix} \text{Re}(z) = \frac{z + \bar{z}}{2} = x \\ \text{Im}(z) = \frac{z - \bar{z}}{2i} = y \end{matrix}$$

- **Multiplication:** Complex numbers are commutative, and distributivity is valid. Be careful: $i^2 = -1$

$$z_1 \cdot z_2 = (x_1 + iy_1) \cdot (x_2 + iy_2) = x_1x_2 + ix_1y_2 + iy_1x_2 + i^2y_1y_2$$

$$= (x_1x_2 - y_1y_2) + i(y_1x_2 + x_1y_2)$$

- **Modulus:** Distance from origin of complex plane

$$\begin{aligned} \sqrt{z\bar{z}} &= \sqrt{x^2 + y^2} \\ |z| &= \sqrt{z\bar{z}} \end{aligned}$$

$$\begin{aligned} z_1 &= x_1 + iy_1 \\ z_2 &= x_2 + iy_2 \end{aligned} \Rightarrow z = \frac{z_1}{z_2}$$

• **Division (ratio):**

- By convention, we don't leave imaginary numbers in the denominator
- Hence, the ratio technique will be to eliminate the imaginary part from the denominator...

$$z = \frac{(x_1 + iy_1)}{(x_2 + iy_2)} = \frac{(x_1 + iy_1)}{(x_2 + iy_2)} \cdot \frac{\overline{z_2}}{\overline{z_2}} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2} = \frac{(x_1x_2 + y_1y_2) + i(y_1x_2 - x_1y_2)}{x_2^2 + y_2^2}$$

○ **Example:**

$$\begin{aligned} \text{if } z_1 &= 1 + 2i \quad \text{find } z = \frac{z_1}{z_2} \\ z_2 &= 2 + 4i \\ z &= \frac{(1 + 2i)}{(2 + 4i)} \cdot \frac{(2 - 4i)}{(2 - 4i)} = \frac{2 - 4i + 4i + 8}{20} = \frac{10}{20} = \frac{1}{2} \end{aligned}$$

• **Triangle inequality:**

$$\hookrightarrow |z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + |z_3| + \dots + |z_n|$$

Lecture 5
ENGR_213

Lecture 5

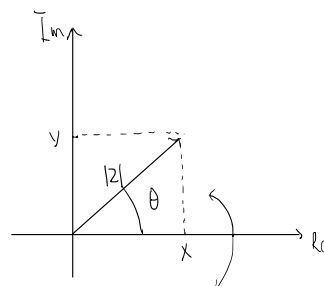
ENGR 213

Chapter 17.2 : Polar Form

Complex numbers
X Degrees ✓ radians

- Similar to vector decomposition in 2 dimension:
- Argument $\theta = \arctan\left(\frac{y}{x}\right)$
- Modulus $|z| = \sqrt{2} = \sqrt{x^2 + y^2}$

$$z = |z|[\cos \theta + i \sin \theta]$$



$z = x + iy$
 $|z| = \text{modulus of } z$
 $\theta = \text{argument of } z$
 $\hookrightarrow \text{pos}(+) \text{ Re axis}$
 $\hookrightarrow \text{principal argument } -\pi < \theta \leq \pi$
 $\hookrightarrow \theta \text{ in rad}$
 $\hookrightarrow \theta = \arctan\left(\frac{y}{x}\right)$

$$Z_1 Z_2 = |Z_1| (\cos \theta_1 + i \sin \theta_1) |Z_2| (\cos \theta_2 + i \sin \theta_2)$$

• Multiplication:

$$Z_1 Z_2 = |Z_1| |Z_2| \left\{ \cos \theta_1 \cos \theta_2 + i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 \right\}$$

$$\approx \quad \left\{ \underbrace{\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2}_{\cos(\theta_1 + \theta_2)} + i \underbrace{(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)}_{\sin(\theta_1 + \theta_2)} \right\}$$

$$Z_1 Z_2 = |Z_1| |Z_2| (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

$$\rightarrow Z = Z_1 Z_2 \Rightarrow |Z| = |Z_1| |Z_2|$$

$$\theta = \theta_1 + \theta_2$$

• Division:

$$Z_1 = |Z_1| (\cos \theta_1 + i \sin \theta_1)$$

$$Z_2 = |Z_2| (\cos \theta_2 + i \sin \theta_2) \quad Z = \frac{Z_1}{Z_2}$$

$$\Rightarrow Z_1 = Z Z_2 \quad |Z_1| = |Z| |Z_2| \rightarrow |Z| = \frac{|Z_1|}{|Z_2|}$$

$$\theta_1 = \theta + \theta_2 \rightarrow \theta = \theta_1 - \theta_2$$

• De Moivre's Formula:

○ Example

• Roots:

- Example: Find the 3 third roots of 1

- EXTRA:

- Euler's Formula
- Cosine
- Sine

Chapter 3.1: Introduction

Definition 25. A n^{th} order linear ordinary differential equation is an equation of the form

$$a_n(x) \frac{d^n}{dx^n} y + a_{n-1}(x) \frac{d^{n-1}}{dx^{n-1}} y + \cdots + a_1(x) \frac{d}{dx} y + a_0(x) y = g(x)$$

Definition 26. An IVP is defined for a n^{th} order linear ODE if

$$y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{n-1}(x_0) = y_{n-1}$$

are defined and known.

General information for the next few definitions:

Hypothesis

All the $a_i(x)$ coefficients are continuous functions.

The function $g(x)$ is a continuous function (0 is considered continuous for homogeneous equations).

The value of $a_n(x) \neq 0$ for all x in the studied interval for any n -order linear equation.

Definition 28. A set of function $f_1(x), f_2(x), \dots, f_n(x)$ is **linearly dependent** on an interval I if there exist some C_i not all zero such that

$$C_1 f_1(x) + C_2 f_2(x) + \dots + C_n f_n(x) = 0$$

for all x in I .

If the set is not linearly dependent, it is **linearly independent**.

Definition 30. Suppose each of the function of a set $f_1, f_2, \dots, f_n$ have at least $n - 1$ derivative. The determinant

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix} \quad W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{n-1} & f_2^{n-1} & \cdots & f_n^{n-1} \end{vmatrix}$$

is called the **Wronskian**.

$$\det(A) = \begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} = a_{1,1}(a_{2,2}a_{3,3} - a_{2,3}a_{3,2}) - a_{1,2}(a_{2,1}a_{3,3} - a_{2,3}a_{3,1}) + a_{1,3}(a_{2,1}a_{3,2} - a_{2,2}a_{3,1})$$

Theorem 31. A set of function $y_1, \dots, y_n$ solutions of a n^{th} order homogeneous linear differential equation on some interval I is linearly independent on I if and only if

$$W(y_1, y_2, \dots, y_n) \neq 0$$

If an n^{th} order linear ODE has n non-trivial linearly independent solution on an interval I , then the set of solution is called a **fundamental set of solution**.

Theorem 32. Let $y_1, y_2, \dots, y_n$ be solutions of an homogeneous n^{th} order differential equation on some interval I . The linear combination of the solutions,

$$y = c_1y_1 + c_2y_2 + \cdots + c_ky_k$$

where the c_i values are constant, is also a solution. **Superposition theorem**

Corollary 33. A constant solution of a solution is a solution.

For homogeneous equation, the trivial solution is a solution.

Theorem 34. *If the set $y_1, y_2, \dots, y_n$ is a fundamental set of solutions for a given ODE on some interval I , then*

$$y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

*is the **general solution** of the homogeneous linear n^{th} order equation.*